

Professional Experience

- Assistant Professor** 2025 - pres.
Mechanical and Industrial Engineering
University of Illinois at Chicago
- Affiliate Member** 2025 - pres.
NSF-Simons National Institute for Theory and Mathematics in Biology
- Postdoctoral Scholar** 2021 - 2025.
University of Chicago
Advisor: Vincenzo Vitelli

Education

- Ph.D. in Physics** 2015-2021
Yale University
Advisor: Michael P. Murrell
Thesis: Energetics of biological mechanics and dynamics
- B.A. in Individualized Study** 2010-2014
New York University

Academic Publications

13. **The four-fold way to rupture in active solids.** Daniel S. Seara, Vincenzo Vitelli. [Nature Materials](#) (2026)
12. **When is nonreciprocity relevant?** Giulia Garcia Lorenzana, David G. Martin, Yael Avni, Daniel S. Seara, Michel Fruchart, Giulio Biroli, and Vincenzo Vitelli. [arXiv:2509.17972](#) (2025)
11. **Information theory for data-driven model reduction in physics and biology.** Matthew S. Schmitt, Maciej Koch-Janusz, Michel Fruchart, Daniel S. Seara, Michael Rust, and Vincenzo Vitelli. [arXiv:2312.06608](#) (2023)
10. **The transition to collective motion in nonreciprocal active matter: coarse graining agent-based models into fluctuating hydrodynamics.** David G. Martin, Daniel S. Seara, Yael Avni, Michel Fruchart, and Vincenzo Vitelli. [PRX](#) (2025), [arXiv:2307.08251](#)
9. **Sociohydrodynamics: data-driven modeling of social behavior.** Daniel S. Seara, Jonathan Colen, Michel Fruchart, Yael Avni, David G. Martin, and Vincenzo Vitelli. [PNAS](#) (2025), [arXiv:2312.17627](#).
8. **Nonreciprocal Ising Model.** Yael Avni, Michel Fruchart, David G. Martin, Daniel S. Seara, and Vincenzo Vitelli. [Physical Review Letters](#) (2025), [arXiv:2311.05471](#)
7. **Dynamical phase transitions in the non-reciprocal Ising model.** Yael Avni, Michel Fruchart, David G. Martin, Daniel S. Seara, and Vincenzo Vitelli. [Physical Review E](#) (2025), [arXiv:2409.07481](#)

6. **Energy partitioning in the cell cortex.** Sheng Chen, Daniel S. Seara, Ani Michaud, Songeun Kim, Willam Bement, and Michael P. Murrell. [Nature Physics](#) (2024), [bioRxiv](#)
5. **Non-reciprocal interactions spatially propagate fluctuations in a 2D Ising model.** Daniel S. Seara, Akash Piya, and A. Pasha Tabatabai. [J. Stat. Mech.](#) (2023), [arXiv:2210.11229](#)
4. **Irreversibility in dynamical phases and transitions.** Daniel S. Seara, Benjamin B. Machta, Michael P. Murrell. [Nature Communications](#) (2021), [arXiv:1911.10696](#)
3. **Detailed balance broken by catch bond kinetics enables mechanical adaptation in active materials.** A. Pasha Tabatabai*, Daniel S. Seara*, Joseph Tibbs, Ian Linsmeier, and Michael P. Murrell. [Advanced Functional Materials](#) (2020)
2. **Wound healing coordinates actin architectures to regulate mechanical work.** Visar Ajeti, A. Pasha Tabatabai, Andrew J. Fleszar, Michael F. Staddon, Daniel S. Seara, Cristian Suarez, M. Sulaiman Yousafzai, Dapeng Bi, David R. Kovar, Shiladitya Banerjee, and Michael P. Murrell. [Nature Physics](#) (2019)
1. **Entropy production rate is maximized in non-contractile actomyosin.** Daniel S. Seara, Vikrant Yadav, Ian Linsmeier, A. Pasha Tabatabai, Patrick W. Oakes, S. M. Ali Tabei, Shiladitya Banerjee, and Michael P. Murrell. [Nature Communications](#) (2018), [arXiv:1804.04232](#)

Talks

Invited

Physics Department Seminar, <i>Oregon State University, USA</i>	May 2026
Institute for Fundamental Science Seminar, <i>U Oregon, USA</i>	May 2026
Applied and Computational Mathematics Seminar, <i>UW Madison</i>	Apr 2026
ISE Graduate Seminar, <i>UIUC, USA</i>	Mar 2026
Nanoscale Materials Theory Group Seminar, <i>Argonne National Lab, USA</i>	Feb 2026
Center for Soft Matter Research Seminar, <i>NYU, USA</i>	Feb 2026
Levitated Matter Workshop, <i>UChicago, USA</i>	Oct 2025
Physics Department Biophysics Seminar, <i>UIC, USA</i>	Sep 2025
Civil and Environmental Engineering Seminar, <i>UMinnesota, USA</i>	Feb 2025
Condensed Matter Seminar, <i>UMass Amherst, USA</i>	Feb 2025
EconophysiX Seminar, <i>Capital Fund Management, France</i>	Feb 2024
Biological Physics Seminar, <i>MPI PKS, Germany</i>	Jan 2024
Quantitative Life Sciences Seminar, <i>ICTP, Italy</i>	Jan 2021
MRSEC Rising Stars in Soft and Biological Matter Symposium, <i>UChicago, USA</i>	Oct 2020
Gotham-Metro Condensed Matter Conference, <i>NYU, USA</i>	Oct 2019

Contributed

LIV Winter Meeting on Statistical Physics, <i>Monterrey, Mexico</i>	Jan 2026
NITMB model reduction workshop, <i>Chicago, USA</i>	Feb 2025
APS March Meeting, <i>Minneapolis, USA</i>	Mar 2024
Journées de Physique Statistique, <i>Paris, France</i>	Jan 2024
STATPHYS28, <i>Tokyo, Japan</i>	Aug 2023
APS March Meeting, <i>online</i>	Mar 2021
CMD2020GEFES, <i>Barcelona, Spain</i>	Aug 2020
APS March Meeting, <i>Denver, USA</i>	Mar 2020
APS March Meeting, <i>Boston, USA</i>	Mar 2019
APS March Meeting, <i>Los Angeles, USA</i>	Mar 2018

Awards

Kadanoff-Rice Postdoctoral Fellowship, University of Chicago	2021
Finalist for Schmidt Science Fellowship	2021
UChicago "Rising Stars in Soft and Biological Matter"	2020
DBIO Travel Grant to 2020 APS March Meeting	2020
NSF GRFP Fellowship	2016
Richard J. Koppenaal Award for Distinguished Interdisciplinary Study	2014
Physics National Honor Society ($\Sigma\Pi\Sigma$)	2013

Service

Professional society membership

American Physical Society (DSNP, DBIO, DSOF)	2017 - pres.
Society for Industrial and Applied Mathematics	2025 - pres.

Organizing

APS Short course on "Physics of Human Social Systems"	Mar 2025
APS session on "Physics of Human Social Systems", DSNP	Mar 2025

Peer review

Physical Review X, PRX Life, Physical Review E, Physical Review Research, Annual Reviews of Condensed Matter Physics, Nature Materials, Experiments in Fluids.

Teaching

University of Illinois Chicago

ME 210 — Engineering Dynamics	Spring 2026
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Yale University (Teaching assistant)

Introduction to Biomechanics (<i>undergraduate and graduate, guest lecturer</i>)	Fall 2017
Statistical Mechanics (<i>graduate</i>)	Spring 2017
Intro Physics for the Life Sciences I	Fall 2016
Intro Physics for the Life Sciences II	Spring 2016
Intro Physics for the Life Sciences I	Fall 2015

New York University (Teaching Assistant)

Physics III	Spring 2015
Classical & Quantum Waves	Fall 2014

Popular Science Publications

APS March Meeting self-organizes online , SoftBites	Mar 2020
Spell-checking biology , SoftBites	Oct 2018
Scaling up biology , SoftBites	Aug 2018
Metastasis: a force to be reckoned with , Hartford Courant	May 2018
How the leopard got its spots , SoftBites	Apr 2018
Dividing liquid droplets as protocells , SoftBites	Feb 2018

Editor for >10 articles for [SoftBites](#)

Outreach

2020 Conference for undergraduate women in physics at Yale organizer	2019
Pathways Summer Scholars Instructor	2018
Splash at Yale Instructor	2017
Yale Science Diplomats	2015-2018
Girl's Science Investigation	2015-2020
"Science as a Creative Endeavor", TedXGallatin lecture	2014
Camp New Day	2013-2015
Co-director of summer science camp for Masa-MexEd	2012

References

- **Vincenzo Vitelli** (Post-doc advisor)
Professor of Physics, University of Chicago
vitelli@uchicago.edu
- **Michael P. Murrell** (Ph.D. advisor)
Associate Professor of Biomedical Engineering & Physics, Yale University
michael.murrell@yale.edu
- **Benjamin B. Machta**
Associate Professor of Physics, Yale University
benjamin.machta@yale.edu